Case Report

Cryptogenic *Burkholderia cepacia* Sepsis in A Haemoglobin E Disease Patient with Splenic and Hepatic Abscesses

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Abstract

We are reporting an unusual case of *Burkholderiacepacia*sepsis in a Hemoglobin E disease patient. The patient had sepsis with both hepatic and splenic abscesses; but unlike the usual scenario, it does not have any identifiable established initial portal of entry including pneumonia. Such a patient, till date, has never been reported in standard literatures.

INTRODUCTION :

We report a case of *B. cepacia* sepsis with hepatic and splenic abscesses in a haemoglobin E disease patient who had no pneumonia or any other identifi-able source of infection. Such a case has never been reported in standard literatures till date.

CASE REPORT :

A 32-year-old male presented with fever and chills for two months; and cough with expectoration for a month. He did not have any neuro-logical, cardiovascular, gastrointestinal or mus-culoskeletal symptoms. His past medical history revealed multiple hospital admissions for anaemia with frequent blood transfusions.

On examination, he was febrile 101æ%F, heart rate of 126/minute, respiratory rate of 22/min and blood pres-sure of 100/70 mm Hg. He was thin built, poorly nourished and was not in distress. His abdominal examination detected splenomegaly; neurological, respiratory and cardiovascular system ex-aminations were essentially normal.

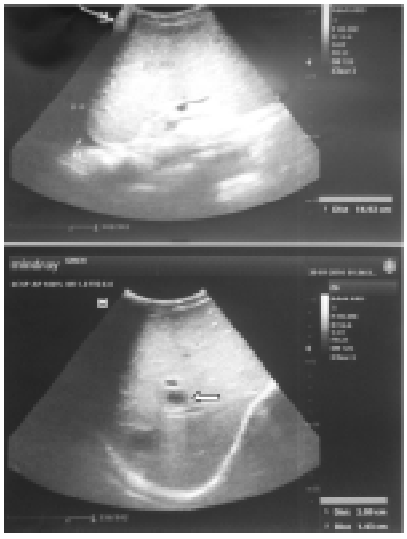
His laboratory tests showed abnormal blood counts; total white cell count was normal, red blood cells were reduced, haemoglobin was 4.1 gm/dl, mean corpuscular volume was 63.9 fl and platelets were reduced at admission. Earlier reports showed a reduced total count (Table 1). The peripheral blood smear showed anisocytosis, microcytosis and target cells. Haemoglobin electrophoresis revealed haemoglobin E homozygous state.

Table 1. Complete Blood Count; temporal profile

	12 days before admission	Day 1	Day 10	Day 14	Day 21
White blood cells (10^3/mm^3)	2.71	5.07	2.51	1.66	2.53
Red blood cells (10^6/mm^3)	4.86	1.80	-	-	-
Hemoglobin (g/dL)	9.6	4.1	3.6	5.7	7.1
Hematocrit (%)	29.4	11.5	11.4	-	-
MCV (10^{-15} L)	60.5	63.9	-	-	-
RDW (%)	50.6	-	-	-	-
Platelet (10^3/mm^3)	113	60	100	55	142

Ultrasonographic examination of the whole abdomen showed an enlarged spleen with heterogenous echotexture and multiple small hypo to anechoic lesions, and a normal sized liver with multiple hypo to anechoic lesions (Figure 1 & 2). A subsequent computed tomography scan of the abdomen revealed

Figure 1: Ultrasound of Liver showing hypoechogenic lesion (arrow)



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Figure 2: USG Spleen showing hypoechogenic lesion (white arrow)



Figure 3: Computed Tomographic Scan showing splenic abscesses (white arrows).



splenomegaly, multiple splenic infarcts with splenic abscesses, hepatomegaly and multiple hepatic abscesses (Figure 3). A guided splenic aspirate was obtained and sent for culture and sensitivity testing.

His urine culture was sterile and X-ray of the chest was normal.

After sending samples for blood culture; patient was empirically started on Piperacillin-Tazobactum and Metronidazole infusions.

On day 3, blood cul-ture showed profuse growth of *B. Cepacia*, sensitive only to Meropenem, Ceftriaxone and Co-trimoxazole. The splenic aspirate too showed growth of *Burkholderia cepacia* sensitive only to Meropenem.

Serology for HIV, Hepatitis B and C, Dengue, Malaria and Salmonella species were negative.

The patient gradually responded to Meropenem injections; started after the culture report. He received 3 units of packed cells transfusion too. His chills and fever subsided, and was subsequently discharged after 20 days in-hospital care.

However he got readmitted again after a few days with renewed fever and weakness and collapsed due to septic shock.

DISCUSSION :

A search of literature did not reveal case reports of *Burkholderia cepacia* sepsis in a patient with haemoglobin E disease with no pneumonia or any other apparent source of infection.

Clinical manifestations of *B. cepacia* infection varies from asymptomatic colonization to necrotizing pneumonia and sepsis[1].

Reports of *B. cepacia* sepsis in patients with cystic fibrosis were plenty [2]. Colonization in cystic fibrosis patients has been reported to be as high as 3% [1].

Some reported catheter-induced *B. cepacia* bacteremia in hemodialysis pa-tients[2] and also in peritoneal dialysis[3].

Immunocompromised patients are at particular risk[4]. There were reports of *B cepacia* sepsis in children with sickle cell disease[5].

There have been reports of pseudo-contamination of blood products by *Burkholderia cepacia* because of the use of contaminated disinfectant during quality control, especially when quarternery ammonium compounds or chlorhexidine were used. With the use of alcohol as disinfectant, it is now rare[6]. Our patient too had multiple transfusions, both during and prior to the current hospital stay, hence making it difficult to identify as the source of infection.

Burkholderia species are often multi-drug-resistant, and treatment is challenging. The organisms are often sensitive to Trimethoprim-Sulfamethoxazole, Meropenem and Ceftazi-dime[7]. The optimal treatment regimen is not established. Intravenously administered antibiotic therapy, often in com-bination, has been successful.

CONCLUSION :

As *B cepacia* is known to cause catastrophic sepsis only in patients with known immunodeficiency; and our patient had solely Hemoglobin E disease as a comorbid

condition; this case report might prove as a good stimulus to study the associated abnormalities in Hemoglobin E disease patients to immunohematolgic cell lines other than red blood cells.

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